

CCNA Mega Lab

Introduction

This lab was sourced from the Jeremy's IT lab YouTube channel and was created by him as a comprehensive resource to practice all the relevant configuration topics for the certification.

The labs topology is as follows:

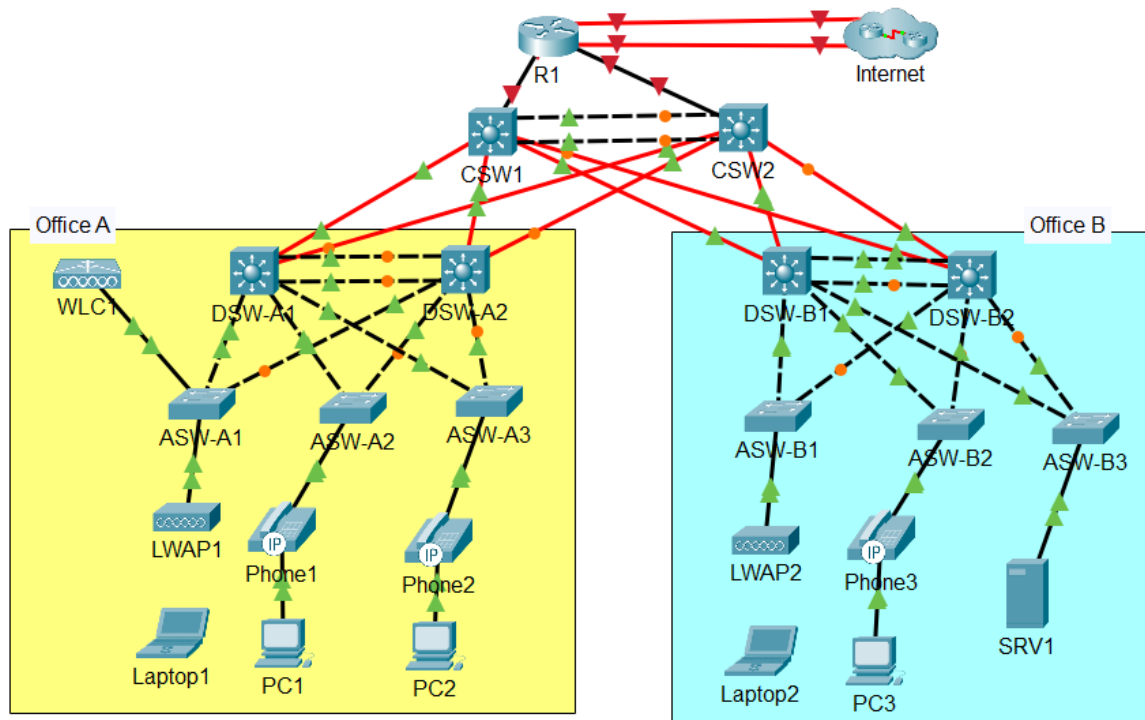


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1.0 Initial Setup

Initial Setup Includes the following steps

1. Configure the appropriate hostname on each router/switch.
2. Configure the enable secret jeremysitlab on each router/switch. Use type 9 hashing if available; otherwise, use type 5.
3. Configure the user account cisco with secret ccna on each router/switch. Use type 9 hashing if available; otherwise, use type 5.
4. Configure the console line to require login with a local user account. Set a 30-minute inactivity timeout. Enable synchronous logging.

1.1 Hostname

This subsection covers the following objective:

Configure the appropriate hostname on each router/switch.

On each router/switch:

Enable

Configure terminal

Hostname <HOSTNAME>

Do write

e.g. for router R1

enable

configure terminal

hostname R1

do write

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#host
Router(config)#hostname R1
R1(config)#Do write
Building configuration...
[OK]
R1(config)#|
```

1.2 Secret

This subsection covers the following objective:

Configure the enable secret jeremysitlab on each router/switch. Use type 9 hashing if available; otherwise, use type 5.

The core/distribution switches support type 9 hashing while the other devices only support type 5

1.2.1 Type 5 Hashing

Enable

Configure Terminal

Enable secret <SECRET>

Do write

e.g. for R1:

enable

configure terminal

enable secret jeremysitlab

do write

1.2.2 Type 9 Hashing

Enable

Configure terminal

Enable algorithm-type scrypt secret <SECRET>

e.g. CSW1

enable

configure terminal

enable algorithm-type scrypt secret jeremysitlab

1.3 User Account

This subsection covers the following objective:

Configure the user account cisco with secret ccna on each router/switch. Use type 9 hashing if available; otherwise, use type 5

Type 5 and type 9 hashing availability is the same as for the section that covers secret configuration

1.3.1 Type 5

Enable

Configure terminal

Username <USERNAME> secret <SECRET>

Do write

e.g. for R1:

enable

configure terminal

username cisco secret ccna

do write

```

R1>en
Password:
R1#
R1#en
R1#
R1#en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#user
R1(config)#username cisco sec
R1(config)#username cisco secret ccna
R1(config)#do wr
Building configuration...
[OK]

R1(config)#do sh ru | i secret
enable secret 5 $1$mERr$LG1PCX0pnG7g1VqYOLG9a/
username cisco secret 5 $1$mERr$Bok4KdfVutXOJolNq009M/

```

The 5 after secret shows the hashing is type 5

1.3.2 Type 9

Enable

Configure terminal

Username <USERNAME> algorithm-type scrypt secret <SECRET>

Do write

e.g. for CSW1

enable

configure terminal

username cisco algorithm-type scrypt secret ccna

do write

```

CSW1>en
Password:
CSW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW1(config)#us
CSW1(config)#username cisco al
CSW1(config)#username cisco algorithm-type s
CSW1(config)#username cisco algorithm-type scrypt s
CSW1(config)#username cisco algorithm-type scrypt secret ccna
CSW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
CSW1(config)#do sh run | i secret
enable secret 9 $9$J19FIAftPZf7c3$13koe1VgzliJg5KQU.JgBhXj4RzTxEUDznF2slb.Hn6
username cisco secret 9 $9$J19FIAftPZf7c3$Mux3wTz1Kth/NxeyEk.E8.QBckaZchVfllsGnx08/0A

```

The 9 after secret shows the hashing is type 9

1.4 Timeout and Synchronous Logging

This subsection covers the following objective:

Configure the console line to require login with a local user account. Set a 30-minute inactivity timeout. Enable synchronous logging.

Enable

jeremysitlab

Configure terminal

Line console 0

Login local

Exec-timeout 30

Logging synchronous

Do write

e.g. R1

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line cons
R1(config)#line console 0
R1(config-line)#log
R1(config-line)#login local
R1(config-line)#exec-
R1(config-line)#exec-timeout 30
R1(config-line)#loggn
R1(config-line)#logg
R1(config-line)#logging syn
R1(config-line)#logging synchronous
R1(config-line)#do write
Building configuration...
[OK]
R1(config-line)#
```

2.0 VLANs and EtherChannel

This part of the lab includes the following steps:

1. In Office A, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-A1 and DSW-A2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel.
2. In Office B, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-B1 and DSW-B2 using an open standard protocol. Both switches should actively try to form an EtherChannel.

3. Configure all links between Access and Distribution switches, including the EtherChannels, as trunk links.
 - a. Explicitly disable DTP on all ports.
 - b. Set each trunk's native VLAN to VLAN 1000 (unused).
 - c. In Office A, allow VLANs 10, 20, 40, and 99 on all trunks.
 - d. In Office B, allow VLANs 10, 20, 30, and 99 on all trunks.
4. Configure one of each office's Distribution switches as a VTPv2 server. Use domain name **JeremysITLab**.
 - a. Verify that other switches join the domain.
 - b. Configure all Access switches as VTP clients.
5. In Office A, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.
 - a. VLAN 10: PCs
 - b. VLAN 20: Phones
 - c. VLAN 40: Wi-Fi
 - d. VLAN 99: Management
6. In Office B, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.
 - a. VLAN 10: PCs
 - b. VLAN 20: Phones
 - c. VLAN 30: Servers
 - d. VLAN 99: Management
7. Configure each Access switch's access port.
 - a. LWAPs will not use FlexConnect
 - b. PCs in VLAN 10, Phones in VLAN 20
 - c. SRV1 in VLAN 30
 - d. Manually configure access mode and explicitly disable DTP
8. Configure ASW-A1's connection to WLC1:
 - a. It must support the Wi-Fi and Management VLANs.
 - b. The Management VLAN should be untagged.
 - c. Disable DTP.
9. Administratively disable all unused ports on Access and Distribution switches.

2.1 Office A EtherChannel

This subsection covers the following objectives:

In Office A, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-A1 and DSW-A2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel.

The Cisco-proprietary protocol is PAgP (Port Aggregation Protocol) and both switches actively trying to form an EtherChannel means both should be in desirable mode.

On DSW-A1 as an example:

First need to find which interfaces are relevant, done with

Enable

Show CDP neighbors

User Access Verification

Username: cisco

Password:

DSW-A1>en

Password:

DSW-A1#sh cdp neigh

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
ASW-A3	Gig 1/0/3	120	S	2960	Gig 0/1
DSW-A2	Gig 1/0/5	120		3650	Gig 1/0/5
DSW-A2	Gig 1/0/4	120		3650	Gig 1/0/4
ASW-A2	Gig 1/0/2	120	S	2960	Gig 0/1
ASW-A1	Gig 1/0/1	120	S	2960	Gig 0/1
CSW1	Gig 1/1/1	120		3650	Gig 1/1/1
CSW2	Gig 1/1/2	120		3650	Gig 1/1/1

DSW-A1#

Relevant interfaces are Gig 1/0/5 and 1/0/4, so following off from checking CDP neighbours:

Configure terminal

Interface range gig 1/0/4-5

Channel-group 1 mode desirable

Do write

DSW-A1#conf t

Enter configuration commands, one per line. End with CNTL/Z.

DSW-A1(config)#int range g1/0/4-5

DSW-A1(config-if-range)#channel-group 1 mode des

DSW-A1(config-if-range)#channel-group 1 mode desirable

DSW-A1(config-if-range)#

Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5, changed state to up

DSW-A1(config-if-range)#do write

Building configuration...

Compressed configuration from 7383 bytes to 3601 bytes[OK]

[OK]

DSW-A1(config-if-range)#

Repeated for DSW-A2

Using do show etherchannel summary we can see that it is in use from the U flag

```
DSW-A1(config-if-range)#do sh etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  S - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

```
Number of channel-groups in use: 1
Number of aggregators:          1
```

Group	Port-channel	Protocol	Ports
1	Po1 (SU)	PAgP	Gig1/0/4 (P) Gig1/0/5 (P)

2.2 Office B EtherChannel

This subsection covers the following objectives:

In Office B, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-B1 and DSW-B2 using an open standard protocol. Both switches should actively try to form an EtherChannel.

The open standard protocol is LACP (Link Aggregation Control Protocol), both should be trying to form an ether channel so both should be in active mode.

Using DSW-B1 as an example:

Recognising correct interfaces

User Access Verification

```
Username: cisco
Password:
```

```
DSW-B1>en
Password:
DSW-B1#show
DSW-B1#show cd
DSW-B1#show cdp ne
DSW-B1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce    Holdtme    Capability    Platform    Port ID
CSW1              Gig 1/1/1          176                3650          Gig 1/1/3
CSW2              Gig 1/1/2          176                3650          Gig 1/1/3
ASW-B1            Gig 1/0/1          176                S             2960          Gig 0/1
DSW-B2            Gig 1/0/4          176                3650          Gig 1/0/4
DSW-B2            Gig 1/0/5          176                3650          Gig 1/0/5
ASW-B2            Gig 1/0/2          176                S             2960          Gig 0/1
ASW-B3            Gig 1/0/3          176                S             2960          Gig 0/1
DSW-B1#
```

Gig 1/0/4 and 1/0/5

From global configuration:

Configure terminal

Interface range gig 1/0/4-5

channel-group 1 mode active

do write

```
DSW-B1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
DSW-B1(config)#int range g1/0/4-5
DSW-B1(config-if-range)#channel-group 1 mode active
DSW-B1(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5, changed state to up

DSW-B1(config-if-range)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Repeated for DSW-B2

Can see from EtherChannel summary that it is in use from the U flag

```
DSW-B2(config-if-range)#do sh etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

Number of channel-groups in use: 1

Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gig1/0/4(P) Gig1/0/5(P)

```
DSW-B2(config-if-range)#
```

2.3 Trunk Links

This subsection covers the following objectives:

Configure all links between Access and Distribution switches, including the EtherChannels, as trunk links.

a. Explicitly disable DTP on all ports.

- b. Set each trunk's native VLAN to VLAN 1000 (unused).
- c. In Office A, allow VLANs 10, 20, 40, and 99 on all trunks.
- d. In Office B, allow VLANs 10, 20, 30, and 99 on all trunks.

2.3.1 Office A

Using DSW-A1 as an example:

First we find the relevant interfaces via

Show cdp neighbors

```
DSW-A1#sh cdp nei
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intfcae  Holdtme    Capability    Platform    Port ID
ASW-A2           Gig 1/0/2        123        S             2960        Gig 0/1
ASW-A1           Gig 1/0/1        123        S             2960        Gig 0/1
CSW1             Gig 1/1/1        123        S             3650        Gig 1/1/1
ASW-A3           Gig 1/0/3        123        S             2960        Gig 0/1
DSW-A2           Por 1            123        S             3650        Gig 1/0/4
DSW-A2           Por 1            123        S             3650        Gig 1/0/5
DSW-A2           Por 1            123        S             3650        Por 1
CSW2             Gig 1/1/2        123        S             3650        Gig 1/1/1
DSW-A1#
```

Gig 1/0/1-3 are the relevant interfaces alongside the portchannels

Interface range gig 1/0/1-3

Switchport mode trunk

Switchport nonegotiate

Switchport trunk native vlan 1000

Switchport trunk allowed vlan 10,20,40,99

```
DSW-A1(config-if-range)#Switchport nonegotiate
DSW-A1(config-if-range)#Switchport trunk native vlan 1000
DSW-A1(config-if-range)#Switchport trunk allowed vlan 10,20,40,99
DSW-A1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up
DSW-A1(config-if-range)#do wr
```

Repeated for the etherchannel, accessed using

Interface port 1

This is then repeated with the access switches

Using ASW-A1 as an example

Show cdp neighbors

```
ASW-A1#sh cdp n
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
DSW-A1         Gig 0/1        147      S           3650      Gig 1/0/1
DSW-A2         Gig 0/2        172      S           3650      Gig 1/0/1
```

Relevant interfaces are g0/1-2

Interface range g0/1-2

Switchport mode trunk

Switchport nonegotiate

Switchport trunk native vlan 1000

Switchport trunk allowed vlan 10,20,40,99

do write

```
ASW-A1(config)#Interface range g0/1-2
ASW-A1(config-if-range)#Switchport mode trunk
```

```
ASW-A1(config-if-range)#Switchport nonegotiate
ASW-A1(config-if-range)#Switchport trunk native vlan 1000
ASW-A1(config-if-range)#Switchport trunk allowed vlan 10,20,40,99
ASW-A1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
do wr
Building configuration...
[OK]
```

2.3.2 Office B

Office B is the exact same with the exception of allowing VLAN 30 instead of 40

Using DSW-B1 as an example:

Show cdp neighbors

```
DSW-B1#sh cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
ASW-B1         Gig 1/0/1        129      S           2960      Gig 0/1
CSW1           Gig 1/1/1        129      S           3650      Gig 1/1/3
DSW-B2         Por 1            129      S           3650      Gig 1/0/4
DSW-B2         Por 1            129      S           3650      Gig 1/0/5
DSW-B2         Por 1            129      S           3650      Por 1
ASW-B2         Gig 1/0/2        129      S           2960      Gig 0/1
ASW-B3         Gig 1/0/3        129      S           2960      Gig 0/1
CSW2           Gig 1/1/2        129      S           3650      Gig 1/1/3
```

Relevant Interfaces are Gig 1/0/1-3

Interface range gig 1/0/1-3

Switchport mode trunk

Switchport nonegotiate

Switchport trunk native vlan 1000

Switchport trunk allowed vlan 10,20,30,99

```
DSW-B1(config)#Interface range gig 1/0/1-3
DSW-B1(config-if-range)#Switchport mode trunk

DSW-B1(config-if-range)#Switchport nonegotiate
DSW-B1(config-if-range)#Switchport trunk native vlan 1000
DSW-B1(config-if-range)#Switchport trunk allowed vlan 10,20,30,99
DSW-B1(config-if-range)#
```

Repeated for interface por 1

```
DSW-B1(config)#int p1
DSW-B1(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/2 (1000),
with ASW-B2 GigabitEthernet0/1 (1).

DSW-B1(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/1 (1000),
with ASW-B1 GigabitEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/3 (1000),
with ASW-B3 GigabitEthernet0/1 (1).

DSW-B1(config-if)#Switchport mode trunk

DSW-B1(config-if)#Switchport nonegotiate
DSW-B1(config-if)#Switchport trunk native vlan 1000
DSW-B1(config-if)#Switchport trunk allowed vlan 10,20,30,99
DSW-B1(config-if)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Need to then repeat for the access switches. Using ASW-B1 as an example:

Show cdp neighbors

```
ASW-B1#sh cdp n
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce  Holdtme    Capability    Platform    Port ID
DSW-B1           Gig 0/1        121        R, S, T, B    3650         Gig 1/0/1
DSW-B2           Gig 0/2        129        R, S, T, B    3650         Gig 1/0/1
```

Relevant ports are g0/1-2

Interface range g0/1-2

Switchport mode trunk

Switchport nonegotiate

Switchport trunk native vlan 1000

Switchport trunk allowed vlan 10,20,30,99

do write


```

ASW-B1(config)#Interface range g0/1-2
ASW-B1(config-if-range)#Switchport mode trunk

ASW-B1(config-if-range)#Switchport nonegotiate
ASW-B1(config-if-range)#Switchport trunk native vlan 1000
ASW-B1(config-if-range)#Switchport trunk allowed vlan 10,20,30,99
ASW-B1(config-if-range)#do write
Building configuration...
[OK]
ASW-B1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

```

2.4 VTPv2 Servers

The objectives of this subsection are:

Configure one of each office's Distribution switches as a VTPv2 server. Use domain name **JeremysITLab**.

- a. Verify that other switches join the domain.
- b. Configure all Access switches as VTP clients.

On DSW-A1:

Enable

Show vtp status (this shows us the default configuration)

```

DSW-A1#sh vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          :
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0001.6395.E700
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 5
Configuration Revision   : 0
MD5 digest               : 0x7D 0x5A 0xA6 0x0E 0x9A 0x72 0xA0 0x3A
                        : 0xF0 0x58 0x10 0x6C 0x9C 0x0F 0xA0 0xF7

```

Configure terminal

Vtp domain JeremysITLab

Vtp version 2

Do write


```
DSW-A1(config)#vtp domain JeremysITLab
Changing VTP domain name from NULL to JeremysITLab
DSW-A1(config)#vtp version 2
DSW-A1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Checking DSW-A2, we can see the changes have propagated

```
DSW-A2#sh vtp stat
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : JeremysITLab
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0001.6492.C600
Configuration last modified by 0.0.0.0 at 2-28-93 15:11:55
Local updater ID is 0.0.0.0 (no valid interface found)
```

The process is the exact same for office B

Configuration done on DSW-B1

```
DSW-B1(config)#Vtp domain JeremysITLab
Changing VTP domain name from NULL to JeremysITLab
DSW-B1(config)#Vtp version 2
DSW-B1(config)#Do write
```

Verifying propagation on DSW-B2

```
DSW-B2#sh vtp status
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : JeremysITLab
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0007.EC8C.4D00
Configuration last modified by 0.0.0.0 at 2-28-93 15:19:42
Local updater ID is 0.0.0.0 (no valid interface found)
```

To make all the access switches VTP clients the following command in configure terminal is used:

Vtp mode client

Before

```
ASW-A1(config)#do sh vtp stat
VTP Version capable          : 1 to 2
VTP version running          : 2
VTP Domain Name              : JeremysITLab
VTP Pruning Mode             : Disabled
VTP Traps Generation         : Disabled
Device ID                    : 0002.4AB5.E200
Configuration last modified by 0.0.0.0 at 2-28-93 15:11:55
Local updater ID is 0.0.0.0 (no valid interface found)
```

Feature VLAN :

VTP Operating Mode : **Server**

After

```
ASW-A1(config)#vtp mode client
Setting device to VTP CLIENT mode.
ASW-A1(config)#do sh vtp status
VTP Version capable          : 1 to 2
VTP version running          : 2
VTP Domain Name              : JeremysITLab
VTP Pruning Mode             : Disabled
VTP Traps Generation         : Disabled
Device ID                    : 0002.4AB5.E200
Configuration last modified by 0.0.0.0 at 2-28-93 15:11:55
```

Feature VLAN :

VTP Operating Mode : **Client**

Repeated for every access switch

2.5 VLAN Configuration/Propagation Via VTP

This subsection includes the following objectives:

In Office A, create and name the following VLANs on one of the Distribution switches.

Ensure that VTP propagates the changes.

- a. VLAN 10: PCs
- b. VLAN 20: Phones
- c. VLAN 40: Wi-Fi
- d. VLAN 99: Management

In Office B, create and name the following VLANs on one of the Distribution switches.

Ensure that VTP propagates the changes.

- a. VLAN 10: PCs
- b. VLAN 20: Phones
- c. VLAN 30: Servers
- d. VLAN 99: Management

2.5.1 Office A

For each VLAN after configure terminal:

Vlan <VLAN NUMBER>

Name <VLAN NAME>

DSW-A1:

```
DSW-A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A1(config)#vlan 10
DSW-A1(config-vlan)#name PCs
DSW-A1(config-vlan)#ex
DSW-A1(config)#vlan 20
DSW-A1(config-vlan)#name Phones
DSW-A1(config-vlan)#ex
DSW-A1(config)#vlan 40
DSW-A1(config-vlan)#name Wi-Fi
DSW-A1(config-vlan)#ex
DSW-A1(config)#vlan 99
DSW-A1(config-vlan)#Management
                        ^
% Invalid input detected at '^' marker.

DSW-A1(config-vlan)#name Management
DSW-A1(config-vlan)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Checking DSW-A2 we can see the changes have propagated

DSW-A2#sh vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9 Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13 Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17 Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21 Gig1/0/22, Gig1/0/23, Gig1/0/24, Gig1/1/1 Gig1/1/2, Gig1/1/3, Gig1/1/4
10	PCs	active	
20	Phones	active	
40	Wi-Fi	active	
99	Management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

2.5.2 Office B

Exact same as Office A except instead of VLAN 40: Wi-Fi we have VLAN 30: Servers

DSW-B1

```

DSW-B1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-B1(config)#vlan 10
DSW-B1(config-vlan)#name PCs
DSW-B1(config-vlan)#ex
DSW-B1(config)#vlan 20
DSW-B1(config-vlan)#name Phones
DSW-B1(config-vlan)#ex
DSW-B1(config)#vlan 30
DSW-B1(config-vlan)#name Servers
DSW-B1(config-vlan)#ex
DSW-B1(config)#vlan 99
DSW-B1(config-vlan)#name Management
DSW-B1(config-vlan)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]

```

Propagated changes on DSW-B2:

```

DSW-B2#sh vlan br

```

VLAN	Name	Status	Ports
1	default	active	Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9 Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13 Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17 Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21 Gig1/0/22, Gig1/0/23, Gig1/0/24, Gig1/1/1 Gig1/1/2, Gig1/1/3, Gig1/1/4
10	PCs	active	
20	Phones	active	
30	Servers	active	
99	Management	active	

2.6 Access Port Configuration

This subsection covers the following objectives:

Configure each Access switch's access port.

- a. Manually configure access mode and explicitly disable DTP
- b. LWAPs will not use FlexConnect
- c. PCs in VLAN 10, Phones in VLAN 20
- d. SRV1 in VLAN 30

part a applies to every switches access port, it is done via:

```

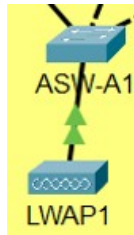
enable
configure terminal
interface <RELEVANT INTERFACE>
switchport mode access
switchport nonegotiate

```

parts b, c and d are specific to switches connected to specific endpoints

2.6.1 LWAPs

The relevant switches here are ASW-A1 and ASW-B1 in this configuration



The LWAP doesn't use CDP so we can't use the show CDP neighbours command to find the correct interface, instead, we take note of its mac address and then use the command

Show mac-address-table

To see which is the relevant interface

Using ASW-A1 as an example:

LWAP mac address

INTERFACE	bandwidth	1000 mbps	100 mbps	10 mbps	Auto
GigabitEthernet0	Duplex	☐	☐	☐	☒
Dot11Radio0	MAC Address	☐	☐	☒	☒

Show mac-address-table

```
ASW-A1#sh mac-address-table
Mac Address Table
-----
```

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
1	0002.4ae7.1601	DYNAMIC	Fa0/1
10	0001.6395.e701	DYNAMIC	Gig0/1
20	0001.6395.e701	DYNAMIC	Gig0/1
40	0001.6395.e701	DYNAMIC	Gig0/1
99	0001.6395.e701	DYNAMIC	Gig0/1

Relevant interface is Fa0/1

Now we have the correct interface we can proceed

Configure terminal

Interface f0/1

Switchport mode access

Switchport nonegotiate

Switchport access vlan 99

```

ASW-A1#Configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A1(config)#Interface f0/1
ASW-A1(config-if)#Switchport mode access
ASW-A1(config-if)#Switchport nonegotiate
ASW-A1(config-if)#Switchport access vlan 99
ASW-A1(config-if)#do write
Building configuration...
[OK]

```

Port	Name	Status	Vlan
Fa0/1		connected	99

ASW-B1:

```

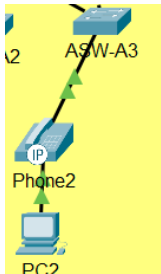
ASW-B1#Configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B1(config)#Interface f0/1
ASW-B1(config-if)#Switchport mode access
ASW-B1(config-if)#Switchport nonegotiate
ASW-B1(config-if)#Switchport access vlan 99
ASW-B1(config-if)#do write
Building configuration...
[OK]

```

99	Management	active	Fa0/1
----	------------	--------	-------

2.6.2 PCs/Phones

Relevant switches are ASW-A2, ASW-A3 and ASW-B2 in the following configuration



Using ASW-A2 as an example

We first find the relevant interface

Show cdp neighbors gives this output

```

ASW-A2#sh cdp neigh
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce  Holdtme    Capability  Platform  Port ID
DSW-A2           Gig 0/2        130        S           3650      Gig 1/0/2
DSW-A1           Gig 0/1        130        S           3650      Gig 1/0/2
Switch           Fas 0/1        129        H P         7960

```

We can see 3 connections, 2 are to the distribution switches and 1 is titled switch

Using show interface status, we can see there are 3 interfaces in use

```

ASW-A2#sh int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Fa0/1     connected   1         auto    auto   10/100BaseTX
Fa0/2     Fa0/2     notconnect  1         auto    auto   10/100BaseTX
Fa0/3     Fa0/3     notconnect  1         auto    auto   10/100BaseTX
Fa0/4     Fa0/4     notconnect  1         auto    auto   10/100BaseTX
Fa0/5     Fa0/5     notconnect  1         auto    auto   10/100BaseTX
Fa0/6     Fa0/6     notconnect  1         auto    auto   10/100BaseTX
Fa0/7     Fa0/7     notconnect  1         auto    auto   10/100BaseTX
Fa0/8     Fa0/8     notconnect  1         auto    auto   10/100BaseTX
Fa0/9     Fa0/9     notconnect  1         auto    auto   10/100BaseTX
Fa0/10    Fa0/10    notconnect  1         auto    auto   10/100BaseTX
Fa0/11    Fa0/11    notconnect  1         auto    auto   10/100BaseTX
Fa0/12    Fa0/12    notconnect  1         auto    auto   10/100BaseTX
Fa0/13    Fa0/13    notconnect  1         auto    auto   10/100BaseTX
Fa0/14    Fa0/14    notconnect  1         auto    auto   10/100BaseTX
Fa0/15    Fa0/15    notconnect  1         auto    auto   10/100BaseTX
Fa0/16    Fa0/16    notconnect  1         auto    auto   10/100BaseTX
Fa0/17    Fa0/17    notconnect  1         auto    auto   10/100BaseTX
Fa0/18    Fa0/18    notconnect  1         auto    auto   10/100BaseTX
Fa0/19    Fa0/19    notconnect  1         auto    auto   10/100BaseTX
Fa0/20    Fa0/20    notconnect  1         auto    auto   10/100BaseTX
Fa0/21    Fa0/21    notconnect  1         auto    auto   10/100BaseTX
Fa0/22    Fa0/22    notconnect  1         auto    auto   10/100BaseTX
Fa0/23    Fa0/23    notconnect  1         auto    auto   10/100BaseTX
Fa0/24    Fa0/24    notconnect  1         auto    auto   10/100BaseTX
Gig0/1    Gig0/1    connected   trunk     auto    auto   10/100BaseTX
Gig0/2    Gig0/2    connected   trunk     auto    auto   10/100BaseTX

```

If 2 are being used for connections to the distribution switch connections (the gig interfaces) then the remaining Fa0/1 must be the one connected to the IP phone. Continuing on:

Configure terminal

Interface f0/1

Switchport mode access

Switchport nonegotiate

Switchport access vlan 10

Switchport voice vlan 20

Do write

```

ASW-A2#Configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A2 (config)#Interface f0/1
ASW-A2 (config-if)#Switchport mode access
ASW-A2 (config-if)#Switchport nonegotiate
ASW-A2 (config-if)#Switchport access vlan 10
ASW-A2 (config-if)#Switchport voice vlan 20
ASW-A2 (config-if)#Do write
Building configuration...
[OK]

```

Repeated exactly for A3 and B2 as they also use fa0/1 as their relevant interface

2.6.3 SRV1

The relevant switch here is ASW-B3



We can easily deduce the relevant interface using the previous approach as from 2.6.2

```
ASW-B3#sh int stat
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Name      connected   1         auto    auto   10/100BaseTX
Fa0/2     notconnect 1         auto    auto   10/100BaseTX
Fa0/3     notconnect 1         auto    auto   10/100BaseTX
Fa0/4     notconnect 1         auto    auto   10/100BaseTX
Fa0/5     notconnect 1         auto    auto   10/100BaseTX
Fa0/6     notconnect 1         auto    auto   10/100BaseTX
Fa0/7     notconnect 1         auto    auto   10/100BaseTX
Fa0/8     notconnect 1         auto    auto   10/100BaseTX
Fa0/9     notconnect 1         auto    auto   10/100BaseTX
Fa0/10    notconnect 1         auto    auto   10/100BaseTX
Fa0/11    notconnect 1         auto    auto   10/100BaseTX
Fa0/12    notconnect 1         auto    auto   10/100BaseTX
Fa0/13    notconnect 1         auto    auto   10/100BaseTX
Fa0/14    notconnect 1         auto    auto   10/100BaseTX
Fa0/15    notconnect 1         auto    auto   10/100BaseTX
Fa0/16    notconnect 1         auto    auto   10/100BaseTX
Fa0/17    notconnect 1         auto    auto   10/100BaseTX
Fa0/18    notconnect 1         auto    auto   10/100BaseTX
Fa0/19    notconnect 1         auto    auto   10/100BaseTX
Fa0/20    notconnect 1         auto    auto   10/100BaseTX
Fa0/21    notconnect 1         auto    auto   10/100BaseTX
Fa0/22    notconnect 1         auto    auto   10/100BaseTX
Fa0/23    notconnect 1         auto    auto   10/100BaseTX
Fa0/24    notconnect 1         auto    auto   10/100BaseTX
Gig0/1    connected  trunk     auto    auto   10/100BaseTX
Gig0/2    connected  trunk     auto    auto   10/100BaseTX
```

```
ASW-B3#sh cdp neigh
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
DSW-B1         Gig 0/1       129     R           3650      Gig 1/0/3
DSW-B2         Gig 0/2       130     R           3650      Gig 1/0/3
```

Fa0/1 is the relevant interface, from here:

Configure terminal

Interface f0/1

Switchport mode access

switchport nonegotiate

switchport access vlan 30

do write

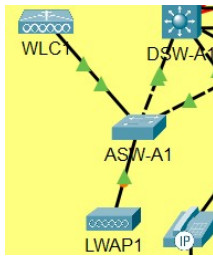
2.7 WLC1

This subsection covers the following objectives:

Configure ASW-A1's connection to WLC1:

- a. It must support the Wi-Fi and Management VLANs.
- b. The Management VLAN should be untagged.
- c. Disable DTP.

This switch was previously configured as it has an LWAP connected



Using show interface status, we get this output

```
ASW-A1#sh int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	99	auto	auto	10/100BaseTX
Fa0/2		connected	1	auto	auto	10/100BaseTX
Fa0/3		notconnect	1	auto	auto	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		notconnect	1	auto	auto	10/100BaseTX
Fa0/24		notconnect	1	auto	auto	10/100BaseTX
Gig0/1		connected	trunk	auto	auto	10/100BaseTX
Gig0/2		connected	trunk	auto	auto	10/100BaseTX

The 2 Gig interfaces are for the distribution switch connection, Fa0/1 is the LWAP connection leaving Fa0/2 as the WLC connection. From here:

Configure terminal

Interface f0/2

Switchport mode trunk

Switchport nonegotiate

Switchport trunk allowed vlan 40,99

Switchport trunk native vlan 99

Do write

```

ASW-A1(config-if)#Switchport mode trunk
ASW-A1(config-if)#Switchport nonegotiate
ASW-A1(config-if)#Switchport trunk allowed vlan 40,99
ASW-A1(config-if)#Switchport trunk native vlan 99
ASW-A1(config-if)#Do write
Building configuration...
[OK]

```

2.8 Disabling Unused Ports

The following subsection covers this topic:

Administratively disable all unused ports on Access and Distribution switches

Using DSW-A1 as an example:

We can check which ports aren't in use with

Show interface status

```

DSW-A1#sh int status

```

Port	Name	Status	Vlan	Duplex	Speed	Type
Pol		connected	trunk	auto	auto	
Gig1/0/1		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/2		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/3		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/4		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/5		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/6		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/7		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/8		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/9		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/10		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/11		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/12		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/13		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/14		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/15		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/16		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/17		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/18		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/19		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/20		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/21		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/22		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/23		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/24		notconnect	1	auto	auto	10/100BaseTX
Gig1/1/1		connected	1	auto	auto	10/100BaseTX
Gig1/1/2		connected	1	auto	auto	10/100BaseTX
Gig1/1/3		notconnect	1	auto	auto	10/100BaseTX
Gig1/1/4		notconnect	1	auto	auto	10/100BaseTX

Every port with the notconnect status needs to be disabled

Done with:

Configure terminal

Interface range g1/0/6-24,g1/1/3-4

Shutdown

Do write

Interface status output shows it was successful

```

DSW-A1(config-if-range)#do sh int status
Port      Name      Status      Vlan      Duplex  Speed Type
Po1                connected   trunk     auto     auto
Gig1/0/1          connected   trunk     auto     auto    10/100BaseTX
Gig1/0/2          connected   trunk     auto     auto    10/100BaseTX
Gig1/0/3          connected   trunk     auto     auto    10/100BaseTX
Gig1/0/4          connected   trunk     auto     auto    10/100BaseTX
Gig1/0/5          connected   trunk     auto     auto    10/100BaseTX
Gig1/0/6          disabled 1         auto     auto    10/100BaseTX
Gig1/0/7          disabled 1         auto     auto    10/100BaseTX
Gig1/0/8          disabled 1         auto     auto    10/100BaseTX
Gig1/0/9          disabled 1         auto     auto    10/100BaseTX
Gig1/0/10         disabled 1         auto     auto    10/100BaseTX
Gig1/0/11         disabled 1         auto     auto    10/100BaseTX
Gig1/0/12         disabled 1         auto     auto    10/100BaseTX
Gig1/0/13         disabled 1         auto     auto    10/100BaseTX
Gig1/0/14         disabled 1         auto     auto    10/100BaseTX
Gig1/0/15         disabled 1         auto     auto    10/100BaseTX
Gig1/0/16         disabled 1         auto     auto    10/100BaseTX
Gig1/0/17         disabled 1         auto     auto    10/100BaseTX
Gig1/0/18         disabled 1         auto     auto    10/100BaseTX
Gig1/0/19         disabled 1         auto     auto    10/100BaseTX
Gig1/0/20         disabled 1         auto     auto    10/100BaseTX
Gig1/0/21         disabled 1         auto     auto    10/100BaseTX
Gig1/0/22         disabled 1         auto     auto    10/100BaseTX
Gig1/0/23         disabled 1         auto     auto    10/100BaseTX
Gig1/0/24         disabled 1         auto     auto    10/100BaseTX
Gig1/1/1          connected 1         auto     auto    10/100BaseTX
Gig1/1/2          connected 1         auto     auto    10/100BaseTX
Gig1/1/3          disabled 1         auto     auto    10/100BaseTX
Gig1/1/4          disabled 1         auto     auto    10/100BaseTX

```

Process repeated with every switch

Example using an access switch (ASW-A1):

```

Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1                connected   99        auto     auto    10/100BaseTX
Fa0/2                connected   trunk     auto     auto    10/100BaseTX
Fa0/3                notconnect 1         auto     auto    10/100BaseTX
Fa0/4                notconnect 1         auto     auto    10/100BaseTX
Fa0/5                notconnect 1         auto     auto    10/100BaseTX
Fa0/6                notconnect 1         auto     auto    10/100BaseTX
Fa0/7                notconnect 1         auto     auto    10/100BaseTX
Fa0/8                notconnect 1         auto     auto    10/100BaseTX
Fa0/9                notconnect 1         auto     auto    10/100BaseTX
Fa0/10               notconnect 1         auto     auto    10/100BaseTX
Fa0/11               notconnect 1         auto     auto    10/100BaseTX
Fa0/12               notconnect 1         auto     auto    10/100BaseTX
Fa0/13               notconnect 1         auto     auto    10/100BaseTX
Fa0/14               notconnect 1         auto     auto    10/100BaseTX
Fa0/15               notconnect 1         auto     auto    10/100BaseTX
Fa0/16               notconnect 1         auto     auto    10/100BaseTX
Fa0/17               notconnect 1         auto     auto    10/100BaseTX
Fa0/18               notconnect 1         auto     auto    10/100BaseTX
Fa0/19               notconnect 1         auto     auto    10/100BaseTX
Fa0/20               notconnect 1         auto     auto    10/100BaseTX
Fa0/21               notconnect 1         auto     auto    10/100BaseTX
Fa0/22               notconnect 1         auto     auto    10/100BaseTX
Fa0/23               notconnect 1         auto     auto    10/100BaseTX
Fa0/24               notconnect 1         auto     auto    10/100BaseTX
Gig0/1              connected   trunk     auto     auto    10/100BaseTX
Gig0/2              connected   trunk     auto     auto    10/100BaseTX

```

Configure terminal

Interface range f0/3-24

Shutdown

Do write

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	99	auto	auto	10/100BaseTX
Fa0/2		connected	trunk	auto	auto	10/100BaseTX
Fa0/3		disabled 1		auto	auto	10/100BaseTX
Fa0/4		disabled 1		auto	auto	10/100BaseTX
Fa0/5		disabled 1		auto	auto	10/100BaseTX
Fa0/6		disabled 1		auto	auto	10/100BaseTX
Fa0/7		disabled 1		auto	auto	10/100BaseTX
Fa0/8		disabled 1		auto	auto	10/100BaseTX
Fa0/9		disabled 1		auto	auto	10/100BaseTX
Fa0/10		disabled 1		auto	auto	10/100BaseTX
Fa0/11		disabled 1		auto	auto	10/100BaseTX
Fa0/12		disabled 1		auto	auto	10/100BaseTX
Fa0/13		disabled 1		auto	auto	10/100BaseTX
Fa0/14		disabled 1		auto	auto	10/100BaseTX
Fa0/15		disabled 1		auto	auto	10/100BaseTX
Fa0/16		disabled 1		auto	auto	10/100BaseTX
Fa0/17		disabled 1		auto	auto	10/100BaseTX
Fa0/18		disabled 1		auto	auto	10/100BaseTX
Fa0/19		disabled 1		auto	auto	10/100BaseTX
Fa0/20		disabled 1		auto	auto	10/100BaseTX
Fa0/21		disabled 1		auto	auto	10/100BaseTX
Fa0/22		disabled 1		auto	auto	10/100BaseTX
Fa0/23		disabled 1		auto	auto	10/100BaseTX
Fa0/24		disabled 1		auto	auto	10/100BaseTX
Gig0/1		connected	trunk	auto	auto	10/100BaseTX
Gig0/2		connected	trunk	auto	auto	10/100BaseTX

ASW-A1 has 4 connection vs every other access switches 3 so the interface range is different, including 1 extra interface

Configure terminal

Interface range f0/2-24

Shutdown

Do write

3.0 Layer 3

This part of the lab includes the following steps (reduced here for space):

1. Configure the following IP addresses on R1's interfaces and enable them
2. Enable IPv4 routing on all Core and Distribution switches.
3. Create a Layer-3 EtherChannel between CSW1 and CSW2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel.
4. Configure the following IP addresses on CSW1. Disable all unused interfaces.
5. Configure the following IP addresses on CSW2. Disable all unused interfaces.
6. Configure the following IP addresses on DSW-A1
7. Configure the following IP addresses on DSW-A2
8. Configure the following IP addresses on DSW-B1

9. Configure the following IP addresses on DSW-B2

10. Manually configure SRV1's IP settings

11. Configure the following management IP addresses on the Access switches (interface VLAN 99), and configure the appropriate subnet's first usable address as the default gateway.

12. Configure HSRPv2 group 1 for Office A's Management subnet (VLAN 99). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

13. Configure HSRPv2 group 2 for Office A's PCs subnet (VLAN 10). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

14. Configure HSRPv2 group 3 for Office A's Phones subnet (VLAN 20). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

15. Configure HSRPv2 group 4 for Office A's Wi-Fi subnet (VLAN 40). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

16. Configure HSRPv2 group 1 for Office B's Management subnet (VLAN 99). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

17. Configure HSRPv2 group 2 for Office B's PCs subnet (VLAN 10). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

18. Configure HSRPv2 group 3 for Office B's Phones subnet (VLAN 20). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

19. Configure HSRPv2 group 4 for Office B's Servers subnet (VLAN 30). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

3.1 R1 IP address Configuration

The following subsection covers this sub-step:

1. Configure the following IP addresses on R1's interfaces and enable them:

- a.** G0/0/0: DHCP client
- b.** G0/1/0: DHCP client
- c.** G0/0: 10.0.0.33/30
- d.** G0/1: 10.0.0.37/30
- e.** Loopback0: 10.0.0.76/32

3.1.1 DHCP Clients (a/b)

1. Configure the following IP addresses on R1's interfaces and enable them:

- a. G0/0/0: DHCP client
- b. G0/1/0: DHCP client

DHCP clients will learn their IP addresses via DHCP so a static address is not used

On R1:

enable

configure terminal

interface range g0/0/0, g0/1/0

ip address dhcp

no shutdown

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int range g0/0/0, g0/1/0
R1(config-if-range)#ip
R1(config-if-range)#ip add
R1(config-if-range)#ip address dhcp
R1(config-if-range)#no shut
```

3.1.2 Static IP Addresses (c/d)

1. Configure the following IP addresses on R1's interfaces and enable them:

- c. G0/0: 10.0.0.33/30
- d. G0/1: 10.0.0.37/30

square brackets ([]) indicate 2 different configuration paths

On R1:

enable

configure terminal

interface [g0/0 | g0/1]

ip address [10.0.0.33 255.255.255.252 | 10.0.0.37 255.255.255.252]

no shutdown

```
R1(config)#int g0/0
R1(config-if)#ip addr 10.0.0.33 255.255.255.252
R1(config-if)#no shut
R1(config)#int g0/1
R1(config-if)#ip addr 10.0.0.37 255.255.255.252
R1(config-if)#no shut
```

3.1.3 Loopback (e)

1. Configure the following IP addresses on R1's interfaces and enable them:

e. Loopback0: 10.0.0.76/32

On R1:

enable

configure terminal

interface Loopback0

ip address 10.0.0.76 255.255.255.255

```
R1(config)#int Loopback0
```

```
R1(config-if)#
```

```
%LINK-5-CHANGED: Interface Loopback0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
```

```
R1(config-if)#ip addr 10.0.0.76 255.255.255.255
```

then:

do write

to save

3.1.4 Verification

We can then check the changes were successful with:

Show ip interface brief

```
R1(config-if)#do sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.0.33	YES	manual	up	up
GigabitEthernet0/1	10.0.0.37	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
GigabitEthernet0/0/0	203.0.113.2	YES	DHCP	up	up
GigabitEthernet0/1/0	203.0.113.6	YES	DHCP	up	up
Loopback0	10.0.0.76	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

3.2 Enabling IPv4 Routing

The following subsection covers this sub-step:

2. Enable IPv4 routing on all Core and Distribution switches.

Done using the command IP routing on all relevant switches

enable

configure terminal

ip routing

do write

```
CSW1>en
Password:
CSW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW1(config)#ip routing
CSW1(config)#do write
DSW-A1>en
Password:
DSW-A1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A1(config)#ip routing
DSW-A1(config)#do write
```

```
DSW-B1>en
Password:
DSW-B1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-B1(config)#ip routing
DSW-B1(config)#do write
```

3.3 Layer 3 EtherChannel

The following subsection covers this sub-step:

3. Create a Layer-3 EtherChannel between CSW1 and CSW2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel. Configure the following IP addresses:

- a. CSW1 PortChannel1: 10.0.0.41/30
- b. CSW2 PortChannel1: 10.0.0.42/30

first use

show cdp neighbors

to find the relevant interfaces


```
CSW1(config)#do sh cdp n
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intfcae  Holdtme  Capability  Platform  Port ID
DSW-B1         Gig 1/1/3         142      3650        Gig 1/1/1
DSW-A2         Gig 1/1/2         142      3650        Gig 1/1/1
DSW-A1         Gig 1/1/1         142      3650        Gig 1/1/1
DSW-B2         Gig 1/1/4         142      3650        Gig 1/1/1
CSW2           Gig 1/0/2         142      3650        Gig 1/0/2
CSW2           Gig 1/0/3         142      3650        Gig 1/0/3
R1             Gig 1/0/1         142      R           C2900       Gig 0/0
```

the interfaces are G1/0/2 and G1/0/3 on both interfaces

On CSW1 (no switchport turns the switch interface from layer 2 to 3):

```
enable
configure terminal
interface range g1/0/2-3
no switchport
channel-group 1 mode desirable
interface
ip address 10.0.0.41 255.255.255.252
```

```
CSW1(config)#int range g1/0/2-3
CSW1(config-if-range)#no switchport
CSW1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up
CSW1(config-if-range)#channel-group 1 mode desirable
CSW1(config-if-range)#
Creating a port-channel interface Port-channel 1
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up
CSW1(config-if-range)#int Po1
CSW1(config-if)#ip address 10.0.0.41 255.255.255.252
CSW1(config-if)#
```

repeated with CSW2 with the relevant ip address, checking show cdp neighbours:

```
CSW1          Por 1          162          3650          Por 1
```

then show etherchannel group summary

```
CSW2(config-if)#do sh eth sum
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (RU)          PAgP       Gig1/0/2 (P) Gig1/0/3 (P)
```

and then pinging

```
CSW2(config-if)#do ping 10.0.0.41

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.41, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

3.4 CSW1

This subsection covers the following sub-step:

4. Configure the following IP addresses on CSW1. Disable all unused interfaces.

- a. G1/0/1: 10.0.0.34/30
- b. G1/1/1: 10.0.0.45/30
- c. G1/1/2: 10.0.0.49/30
- d. G1/1/3: 10.0.0.53/30
- e. G1/1/4: 10.0.0.57/30
- f. Loopback0: 10.0.0.77/32

First we disable all unused interfaces:

using show interface brief we can see the unused interfaces range from G1/0/4-24

CSW1(config)#do sh int st

Port	Name	Status	Vlan	Duplex	Speed	Type
Pol		connected	routed	auto	auto	
Gig1/0/1		connected	1	auto	auto	10/100BaseTX
Gig1/0/2		connected	routed	auto	auto	10/100BaseTX
Gig1/0/3		connected	routed	auto	auto	10/100BaseTX
Gig1/0/4		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/5		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/6		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/7		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/8		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/9		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/10		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/11		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/12		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/13		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/14		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/15		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/16		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/17		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/18		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/19		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/20		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/21		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/22		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/23		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/24		notconnect	1	auto	auto	10/100BaseTX
Gig1/1/1		connected	1	auto	auto	10/100BaseTX
Gig1/1/2		connected	1	auto	auto	10/100BaseTX
Gig1/1/3		connected	1	auto	auto	10/100BaseTX
Gig1/1/4		connected	1	auto	auto	10/100BaseTX

CSW1(config)#interface range g1/0/4-24

CSW1(config-if-range)#shutdown

Port	Name	Status	Vlan	Duplex	Speed	Type
Pol		connected	routed	auto	auto	
Gig1/0/1		connected	1	auto	auto	10/100BaseTX
Gig1/0/2		connected	routed	auto	auto	10/100BaseTX
Gig1/0/3		connected	routed	auto	auto	10/100BaseTX
Gig1/0/4		disabled 1		auto	auto	10/100BaseTX
Gig1/0/5		disabled 1		auto	auto	10/100BaseTX
Gig1/0/6		disabled 1		auto	auto	10/100BaseTX
Gig1/0/7		disabled 1		auto	auto	10/100BaseTX
Gig1/0/8		disabled 1		auto	auto	10/100BaseTX
Gig1/0/9		disabled 1		auto	auto	10/100BaseTX
Gig1/0/10		disabled 1		auto	auto	10/100BaseTX
Gig1/0/11		disabled 1		auto	auto	10/100BaseTX
Gig1/0/12		disabled 1		auto	auto	10/100BaseTX
Gig1/0/13		disabled 1		auto	auto	10/100BaseTX
Gig1/0/14		disabled 1		auto	auto	10/100BaseTX
Gig1/0/15		disabled 1		auto	auto	10/100BaseTX
Gig1/0/16		disabled 1		auto	auto	10/100BaseTX
Gig1/0/17		disabled 1		auto	auto	10/100BaseTX
Gig1/0/18		disabled 1		auto	auto	10/100BaseTX
Gig1/0/19		disabled 1		auto	auto	10/100BaseTX
Gig1/0/20		disabled 1		auto	auto	10/100BaseTX
Gig1/0/21		disabled 1		auto	auto	10/100BaseTX
Gig1/0/22		disabled 1		auto	auto	10/100BaseTX
Gig1/0/23		disabled 1		auto	auto	10/100BaseTX
Gig1/0/24		disabled 1		auto	auto	10/100BaseTX
Gig1/1/1		connected	1	auto	auto	10/100BaseTX
Gig1/1/2		connected	1	auto	auto	10/100BaseTX
Gig1/1/3		connected	1	auto	auto	10/100BaseTX
Gig1/1/4		connected	1	auto	auto	10/100BaseTX

3.4.1 Static Addresses

4. Configure the following IP addresses on CSW1.

- a. G1/0/1: 10.0.0.34/30
- b. G1/1/1: 10.0.0.45/30
- c. G1/1/2: 10.0.0.49/30
- d. G1/1/3: 10.0.0.53/30
- e. G1/1/4: 10.0.0.57/30

enable

configure terminal

interface [g1/0/1 | g1/1/ [1 | 2 | 3 | 4]]

no switchport

ip address 10.0.0.[34 | 45 | 49 | 53 | 57] 255.255.255.252

```
CSW1(config)#interface g1/0/1
CSW1(config-if)#no switchport
CSW1(config-if)#ip address 10.0.0.34 255.255.255.252
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up
CSW1(config-if)#ip address 10.0.0.34 255.255.255.252
```

3.4.2 Loopback

4. Configure the following IP addresses on CSW1.

- f. Loopback0: 10.0.0.77/32

```
enable
configure terminal
interface loopback0
ip address 10.0.0.77 255.255.255.255
```

3.4.3 Verification

Using show ip interface brief:

```
CSW1(config-if)#do sh ip int br
Interface                               IP-Address      OK? Method Status        Protocol
Port-channel1                          10.0.0.41       YES manual up            up
GigabitEthernet1/0/1                   10.0.0.34       YES manual up            up
GigabitEthernet1/0/2                   unassigned      YES unset up            up
GigabitEthernet1/0/3                   unassigned      YES unset up            up
GigabitEthernet1/0/4                   unassigned      YES unset administratively down down
GigabitEthernet1/0/5                   unassigned      YES unset administratively down down
GigabitEthernet1/0/6                   unassigned      YES unset administratively down down
GigabitEthernet1/0/7                   unassigned      YES unset administratively down down
GigabitEthernet1/0/8                   unassigned      YES unset administratively down down
GigabitEthernet1/0/9                   unassigned      YES unset administratively down down
GigabitEthernet1/0/10                  unassigned      YES unset administratively down down
GigabitEthernet1/0/11                  unassigned      YES unset administratively down down
GigabitEthernet1/0/12                  unassigned      YES unset administratively down down
GigabitEthernet1/0/13                  unassigned      YES unset administratively down down
GigabitEthernet1/0/14                  unassigned      YES unset administratively down down
GigabitEthernet1/0/15                  unassigned      YES unset administratively down down
GigabitEthernet1/0/16                  unassigned      YES unset administratively down down
GigabitEthernet1/0/17                  unassigned      YES unset administratively down down
GigabitEthernet1/0/18                  unassigned      YES unset administratively down down
GigabitEthernet1/0/19                  unassigned      YES unset administratively down down
GigabitEthernet1/0/20                  unassigned      YES unset administratively down down
GigabitEthernet1/0/21                  unassigned      YES unset administratively down down
GigabitEthernet1/0/22                  unassigned      YES unset administratively down down
GigabitEthernet1/0/23                  unassigned      YES unset administratively down down
GigabitEthernet1/0/24                  unassigned      YES unset administratively down down
GigabitEthernet1/1/1                   10.0.0.45       YES manual up            up
GigabitEthernet1/1/2                   10.0.0.49       YES manual up            up
GigabitEthernet1/1/3                   10.0.0.53       YES manual up            up
GigabitEthernet1/1/4                   10.0.0.57       YES manual up            up
Loopback0                             10.0.0.77       YES manual up            up
Vlan1                                  unassigned      YES unset administratively down down
```

3.5 CSW2

The following subsection covers this sub-step:

5. Configure the following IP addresses on CSW2. Disable all unused interfaces.

- a. G1/0/1: 10.0.0.38/30
- b. G1/1/1: 10.0.0.61/30
- c. G1/1/2: 10.0.0.65/30
- d. G1/1/3: 10.0.0.69/30
- e. G1/1/4: 10.0.0.73/30
- f. Loopback0: 10.0.0.78/32

first we find and disable the unused interfaces

show interface brief

Enter configuration commands, one per line. End with CTRL-D.

CSW2(config)#do sh int st

Port	Name	Status	Vlan	Duplex	Speed	Type
Pol		connected	routed	auto	auto	
Gig1/0/1		connected	1	auto	auto	10/100BaseTX
Gig1/0/2		connected	routed	auto	auto	10/100BaseTX
Gig1/0/3		connected	routed	auto	auto	10/100BaseTX
Gig1/0/4		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/5		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/6		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/7		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/8		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/9		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/10		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/11		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/12		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/13		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/14		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/15		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/16		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/17		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/18		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/19		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/20		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/21		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/22		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/23		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/24		notconnect	1	auto	auto	10/100BaseTX
Gig1/1/1		connected	1	auto	auto	10/100BaseTX
Gig1/1/2		connected	1	auto	auto	10/100BaseTX
Gig1/1/3		connected	1	auto	auto	10/100BaseTX
Gig1/1/4		connected	1	auto	auto	10/100BaseTX

interface range g1/0/4-24

shutdown

CSW2(config-if-range)#do sh int st

Port	Name	Status	Vlan	Duplex	Speed	Type
Pol		connected	routed	auto	auto	
Gig1/0/1		connected	1	auto	auto	10/100BaseTX
Gig1/0/2		connected	routed	auto	auto	10/100BaseTX
Gig1/0/3		connected	routed	auto	auto	10/100BaseTX
Gig1/0/4		disabled 1		auto	auto	10/100BaseTX
Gig1/0/5		disabled 1		auto	auto	10/100BaseTX
Gig1/0/6		disabled 1		auto	auto	10/100BaseTX
Gig1/0/7		disabled 1		auto	auto	10/100BaseTX
Gig1/0/8		disabled 1		auto	auto	10/100BaseTX
Gig1/0/9		disabled 1		auto	auto	10/100BaseTX
Gig1/0/10		disabled 1		auto	auto	10/100BaseTX
Gig1/0/11		disabled 1		auto	auto	10/100BaseTX
Gig1/0/12		disabled 1		auto	auto	10/100BaseTX
Gig1/0/13		disabled 1		auto	auto	10/100BaseTX
Gig1/0/14		disabled 1		auto	auto	10/100BaseTX
Gig1/0/15		disabled 1		auto	auto	10/100BaseTX
Gig1/0/16		disabled 1		auto	auto	10/100BaseTX
Gig1/0/17		disabled 1		auto	auto	10/100BaseTX
Gig1/0/18		disabled 1		auto	auto	10/100BaseTX
Gig1/0/19		disabled 1		auto	auto	10/100BaseTX
Gig1/0/20		disabled 1		auto	auto	10/100BaseTX
Gig1/0/21		disabled 1		auto	auto	10/100BaseTX
Gig1/0/22		disabled 1		auto	auto	10/100BaseTX
Gig1/0/23		disabled 1		auto	auto	10/100BaseTX
Gig1/0/24		disabled 1		auto	auto	10/100BaseTX
Gig1/1/1		connected	1	auto	auto	10/100BaseTX
Gig1/1/2		connected	1	auto	auto	10/100BaseTX
Gig1/1/3		connected	1	auto	auto	10/100BaseTX
Gig1/1/4		connected	1	auto	auto	10/100BaseTX

3.5.1 Static Addresses

5. Configure the following IP addresses on CSW2. Disable all unused interfaces.

- a. G1/0/1: 10.0.0.38/30
- b. G1/1/1: 10.0.0.61/30
- c. G1/1/2: 10.0.0.65/30
- d. G1/1/3: 10.0.0.69/30
- e. G1/1/4: 10.0.0.73/30

enable

configure terminal

interface [g1/0/1 | g1/1/ [1 | 2 | 3 | 4]]

no switchport

ip address 10.0.0.[38 | 61 | 65 | 69 | 73] 255.255.255.252

```
CSW2(config)#interface g1/0/1
CSW2(config-if)#no switchport
CSW2(config-if)#ip address 10.0.0.38 255.255.255.252
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up
```

3.5.2 Loopback

5. Configure the following IP addresses on CSW2. Disable all unused interfaces.

- f. Loopback0: 10.0.0.78/32

enable

configure terminal

interface loopback0

ip address 10.0.0.78 255.255.255.255

```
CSW2(config-if)#ip address 10.0.0.78 255.255.255.255
%LINK-5-CHANGED: Interface Loopback0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
```

3.5.3 Verification

Show ip interface brief:

```
CSW2(config-if)#do sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Protocol
Port-channel1	10.0.0.42	YES	manual	up	up
GigabitEthernet1/0/1	10.0.0.38	YES	manual	up	up
GigabitEthernet1/0/2	unassigned	YES	unset	up	up
GigabitEthernet1/0/3	unassigned	YES	unset	up	up
GigabitEthernet1/0/4	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/5	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/6	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/7	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/8	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/9	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/10	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/11	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/12	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/13	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/14	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/15	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/16	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/17	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/18	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/19	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/20	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/21	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/22	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/23	unassigned	YES	unset	administratively down	down
GigabitEthernet1/0/24	unassigned	YES	unset	administratively down	down
GigabitEthernet1/1/1	10.0.0.61	YES	manual	up	up
GigabitEthernet1/1/2	10.0.0.65	YES	manual	up	up
GigabitEthernet1/1/3	10.0.0.69	YES	manual	up	up
GigabitEthernet1/1/4	10.0.0.73	YES	manual	up	up
Loopback0	10.0.0.78	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

3.6 DSW-A1

6. Configure the following IP addresses on DSW-A1:

- a. G1/1/1: 10.0.0.46/30
- b. G1/1/2: 10.0.0.62/30
- c. Loopback0: 10.0.0.79/32

enable

configure terminal

interface [g1/1/1 | g1/1/ 2]

no switchport

ip address 10.0.0.[46 | 62] 255.255.255.252

enable

configure terminal

interface loopback0

ip address 10.0.0.79 255.255.255.255

ip interface brief:


```

DSW-A1(config-if)#do sh ip int br
Interface IP-Address OK? Method Status Protocol
Port-channel1 unassigned YES unset up up
GigabitEthernet1/0/1 unassigned YES unset up up
GigabitEthernet1/0/2 unassigned YES unset up up
GigabitEthernet1/0/3 unassigned YES unset up up
GigabitEthernet1/0/4 unassigned YES unset up up
GigabitEthernet1/0/5 unassigned YES unset up up
GigabitEthernet1/0/6 unassigned YES unset administratively down down
GigabitEthernet1/0/7 unassigned YES unset administratively down down
GigabitEthernet1/0/8 unassigned YES unset administratively down down
GigabitEthernet1/0/9 unassigned YES unset administratively down down
GigabitEthernet1/0/10 unassigned YES unset administratively down down
GigabitEthernet1/0/11 unassigned YES unset administratively down down
GigabitEthernet1/0/12 unassigned YES unset administratively down down
GigabitEthernet1/0/13 unassigned YES unset administratively down down
GigabitEthernet1/0/14 unassigned YES unset administratively down down
GigabitEthernet1/0/15 unassigned YES unset administratively down down
GigabitEthernet1/0/16 unassigned YES unset administratively down down
GigabitEthernet1/0/17 unassigned YES unset administratively down down
GigabitEthernet1/0/18 unassigned YES unset administratively down down
GigabitEthernet1/0/19 unassigned YES unset administratively down down
GigabitEthernet1/0/20 unassigned YES unset administratively down down
GigabitEthernet1/0/21 unassigned YES unset administratively down down
GigabitEthernet1/0/22 unassigned YES unset administratively down down
GigabitEthernet1/0/23 unassigned YES unset administratively down down
GigabitEthernet1/0/24 unassigned YES unset administratively down down
GigabitEthernet1/1/1 10.0.0.46 YES manual up up
GigabitEthernet1/1/2 10.0.0.62 YES manual up up
GigabitEthernet1/1/3 unassigned YES unset administratively down down
GigabitEthernet1/1/4 unassigned YES unset administratively down down
Loopback0 10.0.0.79 YES manual up up
Vlan1 unassigned YES unset administratively down down

```

3.7 DSW-A2

7. Configure the following IP addresses on DSW-A2:

- a. G1/1/1: 10.0.0.50/30
- b. G1/1/2: 10.0.0.66/30
- c. Loopback0: 10.0.0.80/32

enable

configure terminal

interface [g1/1/1 | g1/1/ 2]

no switchport

ip address 10.0.0.[50 | 66] 255.255.255.252

enable

configure terminal

interface loopback0

ip address 10.0.0.80 255.255.255.255

ip interface brief:

```
DSW-A2 (config-if)#do sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
Port-channel1  unassigned      YES unset  up          up
GigabitEthernet1/0/1  unassigned      YES unset  up          up
GigabitEthernet1/0/2  unassigned      YES unset  up          up
GigabitEthernet1/0/3  unassigned      YES unset  up          up
GigabitEthernet1/0/4  unassigned      YES unset  up          up
GigabitEthernet1/0/5  unassigned      YES unset  up          up
GigabitEthernet1/0/6  unassigned      YES unset  administratively down down
GigabitEthernet1/0/7  unassigned      YES unset  administratively down down
GigabitEthernet1/0/8  unassigned      YES unset  administratively down down
GigabitEthernet1/0/9  unassigned      YES unset  administratively down down
GigabitEthernet1/0/10 unassigned      YES unset  administratively down down
GigabitEthernet1/0/11 unassigned      YES unset  administratively down down
GigabitEthernet1/0/12 unassigned      YES unset  administratively down down
GigabitEthernet1/0/13 unassigned      YES unset  administratively down down
GigabitEthernet1/0/14 unassigned      YES unset  administratively down down
GigabitEthernet1/0/15 unassigned      YES unset  administratively down down
GigabitEthernet1/0/16 unassigned      YES unset  administratively down down
GigabitEthernet1/0/17 unassigned      YES unset  administratively down down
GigabitEthernet1/0/18 unassigned      YES unset  administratively down down
GigabitEthernet1/0/19 unassigned      YES unset  administratively down down
GigabitEthernet1/0/20 unassigned      YES unset  administratively down down
GigabitEthernet1/0/21 unassigned      YES unset  administratively down down
GigabitEthernet1/0/22 unassigned      YES unset  administratively down down
GigabitEthernet1/0/23 unassigned      YES unset  administratively down down
GigabitEthernet1/0/24 unassigned      YES unset  administratively down down
GigabitEthernet1/1/1  10.0.0.50       YES manual up          up
GigabitEthernet1/1/2  10.0.0.66       YES manual up          up
GigabitEthernet1/1/3  unassigned      YES unset  administratively down down
GigabitEthernet1/1/4  unassigned      YES unset  administratively down down
Loopback0          10.0.0.80       YES manual up          up
Vlan1              unassigned      YES unset  administratively down down
```

3.8 DSW-B1

8. Configure the following IP addresses on DSW-B1:

- a. G1/1/1: 10.0.0.54/30**
- b. G1/1/2: 10.0.0.70/30**
- c. Loopback0: 10.0.0.81/32**

enable

configure terminal

interface [g1/1/1 | g1/1/ 2]

no switchport

ip address 10.0.0.[54 | 70] 255.255.255.252

enable

configure terminal

interface loopback0

ip address 10.0.0.81 255.255.255.255

ip interface brief:

```

DSW-B1(config-if)#do sh ip int br
Interface IP-Address OK? Method Status Protocol
Port-channel1 unassigned YES unset up up
GigabitEthernet1/0/1 unassigned YES unset up up
GigabitEthernet1/0/2 unassigned YES unset up up
GigabitEthernet1/0/3 unassigned YES unset up up
GigabitEthernet1/0/4 unassigned YES unset up up
GigabitEthernet1/0/5 unassigned YES unset up up
GigabitEthernet1/0/6 unassigned YES unset administratively down down
GigabitEthernet1/0/7 unassigned YES unset administratively down down
GigabitEthernet1/0/8 unassigned YES unset administratively down down
GigabitEthernet1/0/9 unassigned YES unset administratively down down
GigabitEthernet1/0/10 unassigned YES unset administratively down down
GigabitEthernet1/0/11 unassigned YES unset administratively down down
GigabitEthernet1/0/12 unassigned YES unset administratively down down
GigabitEthernet1/0/13 unassigned YES unset administratively down down
GigabitEthernet1/0/14 unassigned YES unset administratively down down
GigabitEthernet1/0/15 unassigned YES unset administratively down down
GigabitEthernet1/0/16 unassigned YES unset administratively down down
GigabitEthernet1/0/17 unassigned YES unset administratively down down
GigabitEthernet1/0/18 unassigned YES unset administratively down down
GigabitEthernet1/0/19 unassigned YES unset administratively down down
GigabitEthernet1/0/20 unassigned YES unset administratively down down
GigabitEthernet1/0/21 unassigned YES unset administratively down down
GigabitEthernet1/0/22 unassigned YES unset administratively down down
GigabitEthernet1/0/23 unassigned YES unset administratively down down
GigabitEthernet1/0/24 unassigned YES unset administratively down down
GigabitEthernet1/1/1 10.0.0.54 YES manual up up
GigabitEthernet1/1/2 10.0.0.70 YES manual up up
GigabitEthernet1/1/3 unassigned YES unset administratively down down
GigabitEthernet1/1/4 unassigned YES unset administratively down down
Loopback0 10.0.0.81 YES manual up up
Vlan1 unassigned YES unset administratively down down

```

3.9 DSW-B1

9. Configure the following IP addresses on DSW-B2:

- a. G1/1/1: 10.0.0.58/30
- b. G1/1/2: 10.0.0.74/30
- c. Loopback0: 10.0.0.82/32

enable

configure terminal

interface [g1/1/1 | g1/1/ 2]

no switchport

ip address 10.0.0.[58 | 74] 255.255.255.252

enable

configure terminal

interface loopback0

ip address 10.0.0.82 255.255.255.255

ip interface brief:

```
DSW-B2(config-if)#do sh ip int br
Interface      IP-Address      OK? Method Status Protocol
Port-channel1  unassigned      YES unset  up        up
GigabitEthernet1/0/1  unassigned      YES unset  up        up
GigabitEthernet1/0/2  unassigned      YES unset  up        up
GigabitEthernet1/0/3  unassigned      YES unset  up        up
GigabitEthernet1/0/4  unassigned      YES unset  up        up
GigabitEthernet1/0/5  unassigned      YES unset  up        up
GigabitEthernet1/0/6  unassigned      YES unset  administratively down down
GigabitEthernet1/0/7  unassigned      YES unset  administratively down down
GigabitEthernet1/0/8  unassigned      YES unset  administratively down down
GigabitEthernet1/0/9  unassigned      YES unset  administratively down down
GigabitEthernet1/0/10 unassigned      YES unset  administratively down down
GigabitEthernet1/0/11 unassigned      YES unset  administratively down down
GigabitEthernet1/0/12 unassigned      YES unset  administratively down down
GigabitEthernet1/0/13 unassigned      YES unset  administratively down down
GigabitEthernet1/0/14 unassigned      YES unset  administratively down down
GigabitEthernet1/0/15 unassigned      YES unset  administratively down down
GigabitEthernet1/0/16 unassigned      YES unset  administratively down down
GigabitEthernet1/0/17 unassigned      YES unset  administratively down down
GigabitEthernet1/0/18 unassigned      YES unset  administratively down down
GigabitEthernet1/0/19 unassigned      YES unset  administratively down down
GigabitEthernet1/0/20 unassigned      YES unset  administratively down down
GigabitEthernet1/0/21 unassigned      YES unset  administratively down down
GigabitEthernet1/0/22 unassigned      YES unset  administratively down down
GigabitEthernet1/0/23 unassigned      YES unset  administratively down down
GigabitEthernet1/0/24 unassigned      YES unset  administratively down down
GigabitEthernet1/1/1  10.0.0.58       YES manual up        up
GigabitEthernet1/1/2  10.0.0.74       YES manual up        up
GigabitEthernet1/1/3  unassigned      YES unset  administratively down down
GigabitEthernet1/1/4  unassigned      YES unset  administratively down down
Loopback0        10.0.0.82       YES manual up        up
Vlan1             unassigned      YES unset  administratively down down
```

3.10 SRV1

The following Subsection covers this sub-step:

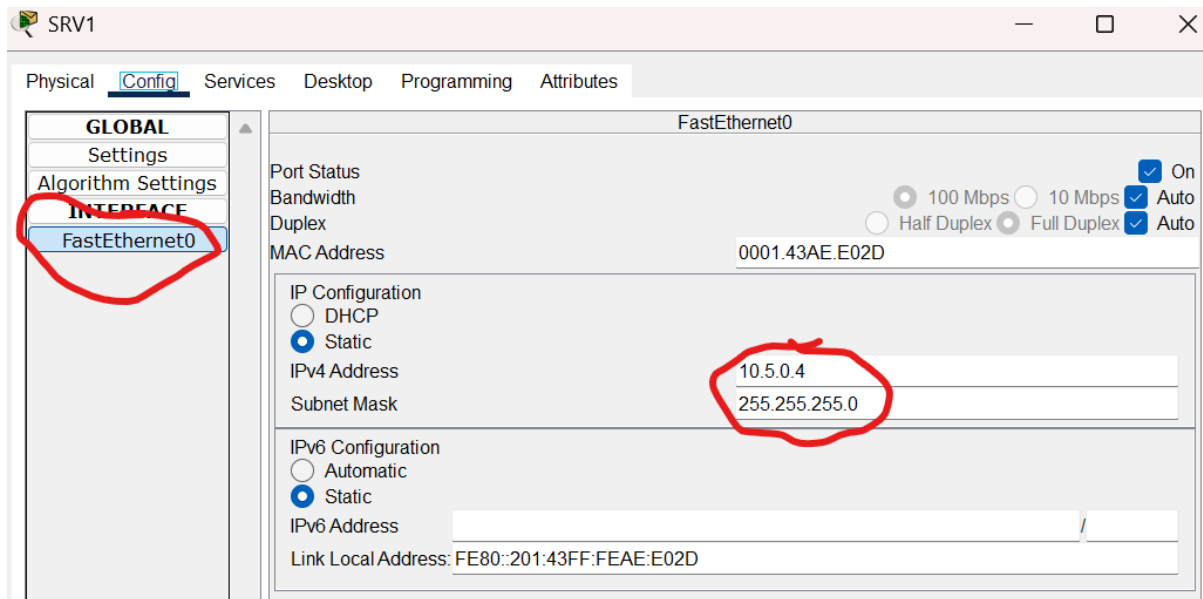
10. Manually configure SRV1's IP settings:

- a. Default Gateway: 10.5.0.1
- b. IPv4 Address: 10.5.0.4
- c. Subnet Mask: 255.255.255.0

For the default gateway:

The screenshot shows the SRV1 configuration window with the 'Config' tab selected. The left sidebar shows the 'INTERFACE' section with 'FastEthernet0' selected. The main area displays 'Global Settings' for the interface. Under 'Gateway/DNS IPv4', the 'Static' radio button is selected, and the 'Default Gateway' field is set to 10.5.0.1. The 'DNS Server' field is empty. The 'Gateway/DNS IPv6' section shows 'Automatic' selected, with empty fields for 'Default Gateway' and 'DNS Server'.

IPv4 Address/Subnet mask:



3.11 Access Switch Management IP Addresses

The following sub-topic covers this sub-step:

11. Configure the following management IP addresses on the Access switches (interface VLAN 99), and configure the appropriate subnet's first usable address as the default gateway.

- a. ASW-A1: 10.0.0.4/28
- b. ASW-A2: 10.0.0.5/28
- c. ASW-A3: 10.0.0.6/28
- d. ASW-B1: 10.0.0.20/28
- e. ASW-B2: 10.0.0.21/28
- f. ASW-B3: 10.0.0.22/28

First useable address is 10.0.0.1 so it acts as the default gateway

enable

configure terminal

ip default-gateway 10.0.0.1

interface vlan999

ip address 10.0.0.[4 | 5 | 6 | 20 | 21 | 22] 255.255.255.240

```
ASW-A1>enable
ASW-A1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A1(config)#ip default-gateway 10.0.0.1
ASW-A1(config)#interface vlan999
ASW-A1(config-if)#ip address 10.0.0.4 255.255.255.240
ASW-A1(config-if)#do wr
Building configuration...
[OK]
```

3.12 HSRP: Office A

The following subsection covers this sub-step:

12. Configure HSRPv2 group 1 for Office A's Management subnet (VLAN 99). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

- a. Subnet: 10.0.0.0/28
- b. VIP: 10.0.0.1
- c. DSW-A1: 10.0.0.2
- d. DSW-A2: 10.0.0.3

13. Configure HSRPv2 group 2 for Office A's PCs subnet (VLAN 10). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

- a. Subnet: 10.1.0.0/24
- b. VIP: 10.1.0.1
- c. DSW-A1: 10.1.0.2
- d. DSW-A2: 10.1.0.3

14. Configure HSRPv2 group 3 for Office A's Phones subnet (VLAN 20). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

- a. Subnet: 10.2.0.0/24
- b. VIP: 10.2.0.1
- c. DSW-A1: 10.2.0.2
- d. DSW-A2: 10.2.0.3

15. Configure HSRPv2 group 4 for Office A's Wi-Fi subnet (VLAN 40). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

- a. Subnet: 10.6.0.0/24
- b. VIP: 10.6.0.1
- c. DSW-A1: 10.6.0.2
- d. DSW-A2: 10.6.0.3

3.12.1 HSRPv2 Group 1 (VLAN 99: Management)

12. Configure HSRPv2 group 1 for Office A's Management subnet (VLAN 99). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

- a. Subnet: 10.0.0.0/28
- b. VIP: 10.0.0.1
- c. DSW-A1: 10.0.0.2
- d. DSW-A2: 10.0.0.3

On DSW-A1:

enable

configure terminal

interface vlan 99

10.0.0.2 255.255.255.240

standby version 2

standby 1 ip 10.0.0.1

standby 1 priority 105

standby 1 preempt

On DSW-A2:

enable

configure terminal

interface vlan 99

10.0.0.3 255.255.255.240

standby version 2

standby 1 ip 10.0.0.1

```
DSW-A1(config)#int vlan 99
DSW-A1(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
DSW-A1(config-if)#
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
DSW-A1(config-if)#
DSW-A1(config-if)#10.0.0.2 255.255.255.240
^
% Invalid input detected at '^' marker.
DSW-A1(config-if)#standby version 2
DSW-A1(config-if)#standby 1 ip 10.0.0.1
% Warning: address is not within a subnet on this interface
DSW-A1(config-if)#standby 1 priority 105
DSW-A1(config-if)#standby 1 preempt
%HSRP-6-STATECHANGE: Vlan99 Grp 1 state Init -> Init
DSW-A1(config-if)#standby 1 preempt
DSW-A2(config)#interface vlan 99
DSW-A2(config-if)#10.0.0.3 255.255.255.240
^
% Invalid input detected at '^' marker.
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 1 ip 10.0.0.1
%LINK-5-CHANGED: Interface Vlan99, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
DSW-A2(config-if)#standby 1 ip 10.0.0.1
% Warning: address is not within a subnet on this interface
DSW-A2(config-if)#
%HSRP-6-STATECHANGE: Vlan99 Grp 1 state Init -> Init
```

3.12.2 HSRPv2 Group 2 (VLAN 10: PCs)

13. Configure HSRPv2 group 2 for Office A's PCs subnet (VLAN 10). Make DSW-A1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A1.

- a. Subnet: 10.1.0.0/24
- b. VIP: 10.1.0.1
- c. DSW-A1: 10.1.0.2
- d. DSW-A2: 10.1.0.3

On DSW-A1:

enable

configure terminal

interface vlan 10

ip address 10.1.0.2 255.255.255.0

standby version 2

standby 2 ip 10.1.0.1

standby 2 priority 105

standby 2 preempt

On DSW-A2:

enable

configure terminal

interface vlan 10

ip address 10.1.0.3 255.255.255.0

standby version 2

standby 2 ip 10.1.0.1

```
DSW-A1(config)#interface vlan 10
DSW-A1(config-if)#10.1.0.2 255.255.255.0
^
% Invalid input detected at '^' marker.

DSW-A1(config-if)#standby version 2
DSW-A1(config-if)#standby 2 ip 10.1.0.1
% Warning: address is not within a subnet on this interface
DSW-A1(config-if)#standby 2 priority 105
DSW-A1(config-if)#standby 2 preempt
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

%HSRP-6-STATECHANGE: Vlan10 Grp 2 state Init -> Init

DSW-A1(config-if)#standby 2 preempt
DSW-A1(config-if)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-A1(config-if)#
%HSRP-6-STATECHANGE: Vlan10 Grp 2 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan10 Grp 2 state Standby -> Active

DSW-A1(config-if)#ip addr 10.1.0.2 255.255.255.0
DSW-A1(config-if)#
```



```

DSW-A2(config)#interface vlan 10
DSW-A2(config-if)#ip address 10.1.0.3 255.255.255.0
DSW-A2(config-if)#standby version 2
DSW-A2(config-if)#standby 2 ip 10.1.0.1
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

DSW-A2(config-if)#standby 2 ip 10.1.0.1
DSW-A2(config-if)#
%HSRP-6-STATECHANGE: Vlan10 Grp 2 state Init -> Init

DSW-A2(config-if)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
DSW-A2(config-if)#

```

3.12.3 HSRPv2 Group 3 (VLAN 20: Phones)

14. Configure HSRPv2 group 3 for Office A's Phones subnet (VLAN 20). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

- a. Subnet: 10.2.0.0/24
- b. VIP: 10.2.0.1
- c. DSW-A1: 10.2.0.2
- d. DSW-A2: 10.2.0.3

On DSW-A2:

enable

configure terminal

interface vlan 20

ip address 10.2.0.3 255.255.255.0

standby version 2

standby 3 ip 10.2.0.1

standby 3 priority 105

standby 3 preempt

On DSW-A1:

enable

configure terminal

interface vlan 20

ip address 10.2.0.2 255.255.255.0

standby version 2

standby 3 ip 10.2.0.2

3.12.3 HSRPv2 Group 4 (VLAN 40: Wi-Fi)

15. Configure HSRPv2 group 4 for Office A's Wi-Fi subnet (VLAN 40). Make DSW-A2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-A2.

- a. Subnet: 10.6.0.0/24
- b. VIP: 10.6.0.1
- c. DSW-A1: 10.6.0.2
- d. DSW-A2: 10.6.0.3

On DSW-A2:

```
enable
configure terminal
interface vlan 40
ip address 10.6.0.3 255.255.255.0
standby version 2
standby 4 ip 10.6.0.1
standby 4 priority 105
standby 4 preempt
```

On DSW-A1:

```
enable
configure terminal
interface vlan 40
ip address 10.6.0.2 255.255.255.0
standby version 2
standby 4 ip 10.6.0.1
```

3.13 HSRP: Office B

The following subsection covers this substep:

16. Configure HSRPv2 group 1 for Office B's Management subnet (VLAN 99). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

- a. Subnet: 10.0.0.16/28
- b. VIP: 10.0.0.17
- c. DSW-B1: 10.0.0.18
- d. DSW-B2: 10.0.0.19

17. Configure HSRPv2 group 2 for Office B's PCs subnet (VLAN 10). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

- a. Subnet: 10.3.0.0/24
- b. VIP: 10.3.0.1
- c. DSW-B1: 10.3.0.2
- d. DSW-B2: 10.3.0.3

18. Configure HSRPv2 group 3 for Office B's Phones subnet (VLAN 20). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

- a. Subnet: 10.4.0.0/24
- b. VIP: 10.4.0.1
- c. DSW-B1: 10.4.0.2
- d. DSW-B2: 10.4.0.3

19. Configure HSRPv2 group 4 for Office B's Servers subnet (VLAN 30). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

- a. Subnet: 10.5.0.0/24
- b. VIP: 10.5.0.1
- c. DSW-B1: 10.5.0.2
- d. DSW-B2: 10.5.0.3

3.13.1 HSRPv2 Group 1 (VLAN 99: Management)

16. Configure HSRPv2 group 1 for Office B's Management subnet (VLAN 99). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

- a. Subnet: 10.0.0.16/28
- b. VIP: 10.0.0.17
- c. DSW-B1: 10.0.0.18
- d. DSW-B2: 10.0.0.19

On DSW-B1:

enable

configure terminal

interface vlan 99

ip address 10.0.0.18 255.255.255.240

standby version 2

standby 1 ip 10.0.0.17

standby 1 priority 105

standby 1 preempt

On DSW-B2:

enable

configure terminal

interface vlan 99

ip address 10.0.0.19 255.255.255.240

standby version 2

standby 1 ip 10.0.0.17

```
DSW-B1(config)#interface vlan 99
DSW-B1(config-if)#ip address 10.0.0.18 255.255.255.240
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 1 ip 10.0.0.17
DSW-B1(config-if)#standby 1 priority 105
DSW-B1(config-if)#standby 1 preempt
%LINK-5-CHANGED: Interface Vlan99, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
```

```
%HSRP-6-STATECHANGE: Vlan99 Grp 1 state Init -> Init
```

```
DSW-B1(config-if)#standby 1 preempt
DSW-B1(config-if)#do wr
```

```
DSW-B2(config)#interface vlan 99
DSW-B2(config-if)#ip address 10.0.0.19 255.255.255.240
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 1 ip 10.0.0.17
%LINK-5-CHANGED: Interface Vlan99, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
```

3.13.2 HSRPv2 Group 2 (VLAN 10: PCs)

17. Configure HSRPv2 group 2 for Office B's PCs subnet (VLAN 10). Make DSW-B1 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B1.

- a. Subnet: 10.3.0.0/24
- b. VIP: 10.3.0.1
- c. DSW-B1: 10.3.0.2
- d. DSW-B2: 10.3.0.3

On DSW-B1:

enable

configure terminal

interface vlan 10

ip address 10.3.0.2 255.255.255.0

standby version 2

standby 2 ip 10.3.0.1

standby 2 priority 105

standby 2 preempt

On DSW-B2:

enable

configure terminal

interface vlan 10

ip address 10.3.0.3 255.255.255.0

standby version 2

standby 2 ip 10.3.0.1

```
DSW-B1(config)#interface vlan 10
DSW-B1(config-if)#ip address 10.3.0.2 255.255.255.0
DSW-B1(config-if)#standby version 2
DSW-B1(config-if)#standby 2 ip 10.3.0.1
DSW-B1(config-if)#standby 2 priority 105
DSW-B1(config-if)#standby 2 preempt
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

%HSRP-6-STATECHANGE: Vlan10 Grp 2 state Init -> Init

DSW-B1(config-if)#standby 2 preempt
DSW-B1(config-if)#do wr
```

```
DSW-B2(config)#interface vlan 10
DSW-B2(config-if)#ip address 10.3.0.3 255.255.255.0
DSW-B2(config-if)#standby version 2
DSW-B2(config-if)#standby 2 ip 10.3.0.1
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
```

3.13.3 HSRPv2 Group 3 (VLAN 20: Phones)

18. Configure HSRPv2 group 3 for Office B's Phones subnet (VLAN 20). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

- a. Subnet: 10.4.0.0/24
- b. VIP: 10.4.0.1
- c. DSW-B1: 10.4.0.2
- d. DSW-B2: 10.4.0.3

On DSW-B2:

```
enable
configure terminal
interface vlan 20
ip address 10.4.0.3 255.255.255.0
standby version 2
standby 3 ip 10.4.0.1
standby 3 priority 105
standby 3 preempt
```

On DSW-B1:

```
enable
configure terminal
interface vlan 20
ip address 10.4.0.2 255.255.255.0
standby version 2
standby 3 ip 10.4.0.1
```

3.13.4 HSRPv2 Group 4 (VLAN 30: Servers)

19. Configure HSRPv2 group 4 for Office B's Servers subnet (VLAN 30). Make DSW-B2 the Active router by increasing its priority to 5 above the default, and enable preemption on DSW-B2.

- a. Subnet: 10.5.0.0/24
- b. VIP: 10.5.0.1
- c. DSW-B1: 10.5.0.2

d. DSW-B2: 10.5.0.3

On DSW-B2:

enable

configure terminal

interface vlan 30

ip address 10.5.0.3 255.255.255.0

standby version 2

standby 4 ip 10.5.0.1

standby 4 priority 105

standby 4 preempt

On DSW-B1:

enable

configure terminal

interface vlan 30

ip address 10.5.0.2 255.255.255.0

standby version 2

standby 4 ip 10.5.0.1

4.0 Spanning Tree Protocol

The following subsection covers this subtopic:

1. Configure Rapid PVST+ on all Access and Distribution switches.

a. Ensure that the Root Bridge for each VLAN aligns with the HSRP Active router by configuring the lowest possible STP priority.

b. Configure the HSRP Standby Router for each VLAN with an STP priority one increment above the lowest priority.

2. Enable PortFast and BPDU Guard on all ports connected to end hosts (including WLC1).

Perform the configurations in interface config mode.

4.1 Rapid PVST+

On each Access/Distribution switch (using DSW-A1 as an example):

Spanning-tree mode rapid-pvst

before (status checked with do show spanning-tree in configure terminal):

```
DSW-A1#show spanning-tree
VLAN0010
    Spanning tree enabled protocol ieee
```

```
VLAN0020
  Spanning tree enabled protocol ieee
```

```
VLAN0040
  Spanning tree enabled protocol ieee
```

```
VLAN0099
  Spanning tree enabled protocol ieee
```

after:

```
VLAN0010
  Spanning tree enabled protocol rstp
```

```
VLAN0020
  Spanning tree enabled protocol rstp
```

```
VLAN0040
  Spanning tree enabled protocol rstp
```

```
VLAN0099
  Spanning tree enabled protocol rstp
```

4.2 Spanning Tree Root Bridge

1. Configure Rapid PVST+ on all Access and Distribution switches.
 - a. Ensure that the Root Bridge for each VLAN aligns with the HSRP Active router by configuring the lowest possible STP priority.
 - b. Configure the HSRP Standby Router for each VLAN with an STP priority one increment above the lowest priority.

4.2.1 Office A

4.2.1.1 Part A

4.2.1.1.1 DSW-A1

Active routers for DSW-A1 are VLANs 99 and 10

On DSW-A1, done with:

```
enable
configure terminal
spanning-tree vlan 99,10 priority 0
```

```

DSW-A1(config)#spanning-tree vlan 99,10 priority 0
DSW-A1(config)#do sh spanning-tree vlan 99
VLAN0099
  Spanning tree enabled protocol rstp
  Root ID    Priority 99
             Address 000C.CFE7.2765
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority 99 (priority 0 sys-id-ext 99)
             Address 000C.CFE7.2765
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Gi1/0/2 Desg FWD 4 128.2 P2p
Gi1/0/1 Desg FWD 4 128.1 P2p
Gi1/0/3 Desg FWD 4 128.3 P2p
Po1 Desg FWD 3 128.29 Shr

DSW-A1(config)#do sh spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority 10
             Address 000C.CFE7.2765
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority 10 (priority 0 sys-id-ext 10)
             Address 000C.CFE7.2765
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Gi1/0/2 Desg FWD 4 128.2 P2p
Gi1/0/1 Desg FWD 4 128.1 P2p
Gi1/0/3 Desg FWD 4 128.3 P2p
Po1 Desg FWD 3 128.29 Shr

```

show spanning-tree vlan [vlan number] can be used to view status

4.2.1.1.2 DSW-A2

DSW-A2 is the active router for VLANs 20 and 40

On DSW-A2, done with (STP increments are in 4096):

enable

configure terminal

spanning-tree vlan 20,40 priority 0


```
DSW-A2(config)#spanning-tree vlan 20,40 priority 0
DSW-A2(config)#do sh spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority 20
             Address 00D0.BC16.1213
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority 20 (priority 0 sys-id-ext 20)
             Address 00D0.BC16.1213
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gil/0/2	Desg	FWD	4	128.2	P2p
Gil/0/1	Desg	FWD	4	128.1	P2p
Gil/0/3	Desg	FWD	4	128.3	P2p
Pol	Altn	BLK	3	128.29	Shr

```
DSW-A2(config)#do sh spanning-tree vlan 40
VLAN0040
  Spanning tree enabled protocol rstp
  Root ID    Priority 40
             Address 00D0.BC16.1213
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority 40 (priority 0 sys-id-ext 40)
             Address 00D0.BC16.1213
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gil/0/2	Desg	FWD	4	128.2	P2p
Gil/0/1	Desg	FWD	4	128.1	P2p
Gil/0/3	Desg	FWD	4	128.3	P2p
Pol	Desg	FWD	3	128.29	Shr

show spanning-tree vlan [vlan number] can be used to view status

4.2.1.2 Part B

4.2.1.2.1 DSW-A1

DSW-A1 is the standby router for VLANs 20 and 40

On DSW-A1, done with (STP increments are in 4096):

enable

configure terminal

spanning-tree vlan 20,40 priority 4096

```
DSW-A1(config)#spanning-tree vlan 20,40 priority 4096
DSW-A1(config)#do sh spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority 20
             Address 00D0.BC16.1213
             Cost 3
             Port 29(Port-channel1)
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority 4116 (priority 4096 sys-id-ext 20)
             Address 000C.CFE7.2765
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Pol	Root	FWD	3	128.29	Shr
Gil/0/3	Desg	FWD	4	128.3	P2p
Gil/0/1	Desg	FWD	4	128.1	P2p
Gil/0/2	Desg	FWD	4	128.2	P2p

```
DSW-A1(config)#do sh spanning-tree vlan 40
VLAN0040
  Spanning tree enabled protocol rstp
  Root ID    Priority 40
             Address 00D0.BC16.1213
             Cost 3
             Port 29(Port-channel1)
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority 4136 (priority 4096 sys-id-ext 40)
             Address 000C.CFE7.2765
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Pol	Root	FWD	3	128.29	Shr
Gil/0/3	Desg	FWD	4	128.3	P2p
Gil/0/1	Desg	FWD	4	128.1	P2p
Gil/0/2	Desg	FWD	4	128.2	P2p

4.2.1.2.2 DSW-A2

DSW-A2 is the standby router for VLANs 20 and 40

On DSW-A2, done with (STP increments are in 4096):

enable

configure terminal

spanning-tree vlan 99,10 priority 4096

```
DSW-A2(config)#do sh span vlan 99
VLAN0099
  Spanning tree enabled protocol rstp
    Root ID    Priority    99
              Address    000C.CFE7.2765
              Cost        3
              Port        29(Port-channel1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID   Priority    4195 (priority 4096 sys-id-ext 99)
              Address    00D0.BC16.1213
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/1        Desg FWD 4         128.1   P2p
Gi1/0/3        Desg FWD 4         128.3   P2p
Gi1/0/2        Desg FWD 4         128.2   P2p
Po1            Root FWD 3         128.29  Shr

DSW-A2(config)#do sh span vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
    Root ID    Priority    10
              Address    000C.CFE7.2765
              Cost        3
              Port        29(Port-channel1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID   Priority    4106 (priority 4096 sys-id-ext 10)
              Address    00D0.BC16.1213
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/1        Desg FWD 4         128.1   P2p
Gi1/0/3        Desg FWD 4         128.3   P2p
Gi1/0/2        Desg FWD 4         128.2   P2p
Po1            Root FWD 3         128.29  Shr
```

4.2.2 Office B

4.2.2.1 Part A

4.2.2.1.1 DSW-B1

DSW-B1 is the active router for VLANs 99 and 10

On DSW-B1, done with:

enable

configure terminal

spanning-tree vlan 99,10 priority 0

```
DSW-B1(config)#spanning-tree vlan 99,10 priority 0
DSW-B1(config)#do sh span vlan 99
VLAN0099
  Spanning tree enabled protocol rstp
  Root ID    Priority    99
             Address    0001.635B.4C6A
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    99 (priority 0 sys-id-ext 99)
             Address    0001.635B.4C6A
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/2      Desg FWD 4         128.2   P2p
Gi1/0/1      Desg FWD 4         128.1   P2p
Gi1/0/3      Desg FWD 4         128.3   P2p
Po1          Desg FWD 3         128.29  Shr

DSW-B1(config)#do sh span vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    10
             Address    0001.635B.4C6A
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    10 (priority 0 sys-id-ext 10)
             Address    0001.635B.4C6A
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/2      Desg FWD 4         128.2   P2p
Gi1/0/1      Desg FWD 4         128.1   P2p
Gi1/0/3      Desg FWD 4         128.3   P2p
Po1          Desg FWD 3         128.29  Shr
```

4.2.2.1.2 DSW-B2

DSW-B2 is the active router for VLANs 20 and 30

On DSW-B1, done with:

enable

configure terminal

spanning-tree vlan 20,30 priority 0

```

DSW-B2(config)#spanning-tree vlan 20,30 priority 0
DSW-B2(config)#do sh span vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    20
             Address     0050.0F7B.3C01
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    20 (priority 0 sys-id-ext 20)
             Address     0050.0F7B.3C01
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Po1          Desg FWD 3         128.29 Shr
Gi1/0/1      Desg FWD 4         128.1  P2p
Gi1/0/3      Desg FWD 4         128.3  P2p
Gi1/0/2      Desg FWD 4         128.2  P2p

DSW-B2(config)#do sh span vlan 30
VLAN0030
  Spanning tree enabled protocol rstp
  Root ID    Priority    30
             Address     0050.0F7B.3C01
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    30 (priority 0 sys-id-ext 30)
             Address     0050.0F7B.3C01
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Po1          Desg FWD 3         128.29 Shr
Gi1/0/1      Desg FWD 4         128.1  P2p
Gi1/0/3      Desg FWD 4         128.3  P2p
Gi1/0/2      Desg FWD 4         128.2  P2p

```

4.2.2.2 Part B

4.2.2.2.1 DSW-B1

DSW-B1 is the active router for VLANs 20 and 30

On DSW-B1, done with:

enable

configure terminal

spanning-tree vlan 20,30 priority 0

```
DSW-B1(config)#spanning-tree vlan 20,30 priority 4096
DSW-B1(config)#do sh span vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    20
            Address    0001.635B.4C6A
            Cost        11
            Port        1(GigabitEthernet1/0/1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4116 (priority 4096 sys-id-ext 20)
            Address    0001.635B.4C6A
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost        Prio.Nbr Type
-----
Gi1/0/2      Altn BLK 4          128.2    P2p
Gi1/0/1      Root FWD 4          128.1    P2p
Gi1/0/3      Desg FWD 4          128.3    P2p
Po1          Desg FWD 3          128.29   Shr

DSW-B1(config)#do sh span vlan 30
VLAN0030
  Spanning tree enabled protocol rstp
  Root ID    Priority    30
            Address    0001.635B.4C6A
            Cost        11
            Port        2(GigabitEthernet1/0/2)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4126 (priority 4096 sys-id-ext 30)
            Address    0001.635B.4C6A
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost        Prio.Nbr Type
-----
Gi1/0/2      Root FWD 4          128.2    P2p
Gi1/0/1      Desg FWD 4          128.1    P2p
Gi1/0/3      Desg FWD 4          128.3    P2p
Po1          Desg FWD 3          128.29   Shr
```

4.2.2.2.2 DSW-B2

DSW-B2 is the passive router for VLANs 99 and 10

On DSW-B2, done with:

enable

configure terminal

spanning-tree vlan 99,10 priority 4096

```
DSW-B2(config)#spanning-tree vlan 99,10 priority 4096
DSW-B2(config)#do sh span vlan 99
VLAN0099
  Spanning tree enabled protocol rstp
  Root ID    Priority    99
            Address    0001.635B.4C6A
            Cost        3
            Port        29(Port-channel1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4195 (priority 4096 sys-id-ext 99)
            Address    0050.0F7B.3C01
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost        Prio.Nbr Type
-----
Po1          Root FWD 3          128.29   Shr
Gi1/0/1      Desg FWD 4          128.1    P2p
Gi1/0/3      Desg FWD 4          128.3    P2p
Gi1/0/2      Desg FWD 4          128.2    P2p

DSW-B2(config)#do sh span vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    10
            Address    0001.635B.4C6A
            Cost        3
            Port        29(Port-channel1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4106 (priority 4096 sys-id-ext 10)
            Address    0050.0F7B.3C01
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost        Prio.Nbr Type
-----
Po1          Root FWD 3          128.29   Shr
Gi1/0/1      Desg FWD 4          128.1    P2p
Gi1/0/3      Desg FWD 4          128.3    P2p
Gi1/0/2      Desg FWD 4          128.2    P2p
```

4.2.3 STP Security Features

2. Enable PortFast and BPDU Guard on all ports connected to end hosts (including WLC1).
Perform the configurations in interface config mode.

Switches with end hosts are all of the access switches

4.2.3.1 ASW-A1

Show cdp neighbors shows which interfaces are connected with distribution switches:

```
ASW-A1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intf     Holdtme    Capability    Platform    Port ID
DSW-A1           Gig 0/1         124        S              3650         Gig 1/0/1
DSW-A2           Gig 0/2         123        S              3650         Gig 1/0/1
```

show interface status gives these 2 remaining interfaces:

```
ASW-A1#sh int status
Port      Name           Status      Vlan      Duplex  Speed  Type
Fa0/1     Gig0/1         connected   99        auto    auto   10/100BaseTX
Fa0/2     Gig0/2         connected   trunk     auto    auto   10/100BaseTX
```

a trunk connection is used for the WLC meaning it is the Fa0/2 interface leaving Fa0/1 for the LWAP

LWAP Interface:

enable

configure terminal

interface f0/1

spanning-tree portfast

spanning-tree bpduguard enable

```
ASW-A1(config)#int f0/1
ASW-A1(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
ASW-A1(config-if)#spanning-tree bpduguard enable
ASW-A1(config-if)#do wr
```

WLC Interface:

enable

configure terminal

interface f0/2

spanning-tree portfast trunk

spanning-tree bpduguard enable

```
ASW-A1(config)#interface f0/2
ASW-A1(config-if)#spanning-tree portfast trunk
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
ASW-A1(config-if)#spanning-tree bpduguard enable
ASW-A1(config-if)#do wr
```

4.2.3.2 All Other Access Switches

All other access switches use Fa0/1 for their endpoint connection and only have one endpoint they are connected to so the same set of commands can be used for all of them

```
enable
configure terminal
interface f0/1
spanning-tree portfast
spanning-tree bpduguard enable
do write
```

ASW-A2 as an example:

```
ASW-A2>enable
ASW-A2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ASW-A2(config)#interface f0/1
ASW-A2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
ASW-A2(config-if)#spanning-tree bpduguard enable
ASW-A2(config-if)#do write
Building configuration...
[OK]
```